Appendix 2. Search strategies

Deep Learning for Inter-Patient ECG Arrhythmia Classification on the MIT-BIH Database: A Systematic Review

All final database searches combined the same five concept blocks with the Boolean operator AND: (i) deep learning, (ii) ECG, (iii) arrhythmia or heartbeat classification, (iv) inter-patient evaluation and (v) MIT-BIH.

PubMed:

```
("deep learning"[tiab] OR "convolutional neural network*"[tiab] OR CNN[tiab] OR LSTM[tiab]
OR transformer*[tiab] OR "neural network*"[tiab])
AND (ECG[tiab] OR electrocardiogra*[tiab] OR heartbeat*[tiab])
AND (arrhythmi*[tiab] OR "heartbeat classification"[tiab] OR "beat classification"[tiab])
AND ("inter-patient"[tiab] OR interpatient[tiab] OR "patient-independent"[tiab] OR "patient
independent"[tiab] OR DS1[tiab])
AND ("MIT-BIH"[tiab] OR MITBIH[tiab])
AND (2020:2026[dp])
```

Scopus:

```
TITLE-ABS-KEY(("deep learning" OR "convolutional neural network*" OR CNN OR LSTM OR
transformer* OR "neural network*") AND (ECG OR electrocardiogra* OR heartbeat*) AND
(arrhythmi* OR "heartbeat classification" OR "beat classification") AND ("inter-patient" OR
interpatient OR "patient-independent" OR "patient independent" OR DS1) AND ("MIT-BIH" OR
MITBIH)) AND PUBYEAR > 2019 AND PUBYEAR < 2027 AND (DOCTYPE(ar) OR DOCTYPE(cp))
```

IEEE Xplore:

```
("deep learning" OR "convolutional neural network" OR CNN OR LSTM OR transformer OR "neural
network") AND (ECG OR electrocardiogram OR heartbeat) AND (arrhythmia OR "heartbeat
classification" OR "beat classification") AND ("inter-patient" OR interpatient OR "patient-
independent" OR "patient independent" OR DS1) AND ("MIT-BIH" OR MITBIH)
```

Filters: publication years 2020-2026.

ACM Digital Library:

```
("deep learning" OR "convolutional neural network" OR CNN OR LSTM OR transformer OR "neural
network") AND (ECG OR electrocardiogram OR heartbeat) AND (arrhythmia OR "heartbeat
classification" OR "beat classification") AND ("inter-patient" OR interpatient OR "patient-
independent" OR "patient independent" OR DS1) AND ("MIT-BIH" OR MITBIH)
```

Searched in: Title, Abstract and Author Keywords. Collection: The ACM Guide to Computing Literature. Filter: publication years 2020-2026.

Google Scholar:

Via Publish or Perish, searched iteratively throughout the review period, last search on 5 October 2026. All of the words: deep learning arrhythmia MIT-BIH. Any of the words: inter-patient interpatient patient-independent. Years 2020-2026. First 200 records by relevance screened. No additional eligible study was found.

Citation searching:

Reference lists of the included studies and of the most recent systematic review on deep learning-based arrhythmia classification.